



Ahsanullah University of Science & Technology

Department of Computer Science & Engineering

Course No : CSE 2214

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Assignment No 3

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Question

1. Write a program that lets the user type some text, consisting of words separated by blanks, ending with a carriage return, and displays each word with its characters reversed, while keeping the order of the words unchanged.

You need to write a procedure to reverse each word.

Answer:

.MODEL SMALL

.STACK 100H

.DATA

MSG_INPUT DB 'Input: \$'

MSG_OUTPUT DB 0DH, 0AH, 'Output: \$'

INPUT_STR DB 200 DUP('\$')

WORD_BUF DB 50 DUP('\$')

.CODE

MAIN PROC

MOV AX, @DATA

MOV DS, AX

LEA DX, MSG_INPUT

MOV AH, 9

INT 21H

LEA SI, INPUT_STR

READ_LINE:

MOV AH, 1

INT 21H

CMP AL, 0DH

JE READ_DONE

MOV [SI], AL

INC SI

JMP READ_LINE

READ_DONE:

MOV BYTE PTR [SI], '\$'

LEA DX, MSG_OUTPUT

MOV AH, 9
INT 21H

LEA SI, INPUT_STR
LEA DI, WORD_BUF
XOR CX, CX

PROCESS_LOOP:

MOV AL, [SI]
CMP AL, '\$'
JE LAST_WORD

CMP AL, ''
JE HANDLE_SPACE

MOV [DI], AL
INC SI
INC DI
INC CX
JMP PROCESS_LOOP

HANDLE_SPACE:

CALL REVERSE_AND_PRINT

MOV DL, ''
MOV AH, 2
INT 21H

LEA DI, WORD_BUF
XOR CX, CX
INC SI
JMP PROCESS_LOOP

LAST_WORD:

CALL REVERSE_AND_PRINT

MOV AH, 4CH
INT 21H

MAIN ENDP

REVERSE_AND_PRINT PROC

JCXZ DONE_REV

PUSH SI
LEA SI, WORD_BUF
ADD SI, CX
DEC SI

PRINT_LOOP:
 MOV DL, [SI]
 MOV AH, 2
 INT 21H
 DEC SI
 LOOP PRINT_LOOP

POP SI

DONE_REV:
 RET

REVERSE_AND_PRINT ENDP

END MAIN